

# PARAM BIYANI

 prmbiy.github.io    parambiyani8@gmail.com  +91 8425935760  Google Scholar

## EDUCATION

**BITS Pilani - K.K. Birla Goa Campus**

*B.E. in Computer Science, Minor in Data Science*

*August 2019 - July 2023*

**CGPA: 8.96/10**

## WORK EXPERIENCE

**Machine Alignment, Transparency, and Security Program | MATS AI Safety** |  

London, UK

*Research Fellow; Working with Dr. Krishnamurthy (Dj) Dvijotham (Google DeepMind)*

*June 2026 - present*

- Engineered a scalable pre-flight verification system to detect broken or impossible tasks before agents attempt them, directly mitigating reward hacking and deceptive behaviors. Able to detect 89% and prove 67% of impossible tasks in SWE-Bench.
- Developed a formalization pipeline in Lean 4 that translates task intent and proves if spec and tests are incompatible.

**Proximal** | 

Sept 2026 - present

- Working on building reward hack safe harnesses and monitoring systems for frontier LLM RL training.
- Working on developing alignment evals and benchmarks. In collaboration with MATS.

**Supervised Program for Alignment Research | SPAR AI Safety** | 

Remote

*Research Fellow*

*February 2026 - June 2026*

- Developing a method to evaluate agent shutdownability and assess the extent to which agents exhibit evaluation awareness.
- Investigating when agents fail to comply with shutdown and developing mitigation strategies to improve reliability.

**Qualcomm**

Hyderabad, India

*Senior AI Engineer*

*December 2025 - June 2026*

- Designed an automated NL to spec generator (autoformalization) for Qualcomm's chip design verification pipelines.
- Developed an autonomous coding agent for C and Verilog, used by 15k+ developers and commits 1M+ lines of code daily.

**Microsoft, PROSE Team**

Bangalore, India

*Research Fellow*

*October 2023 - October 2025*

- Led research on an autonomous coding agents; contributions shipped to **GitHub Copilot**, with a 15% gain on swe-bench-verified over the baseline swe-agent.
- Improved agentic performance across long term interaction benchmarks (SpreadSheetBench,  $\tau$ -bench, OSWorld (autonomous computer-use)) tasks by 10–20% using a novel bootstrapping algorithm that learns knowledge bases from past experiences.
- Developed a system to autoformalize informal olympiad math problems to Lean 4 theorems, achieving a 95% automation.
- Designed a rubric-based learning algorithm to infer human satisfaction and task relevance from conversational data, deployed across **Visual Studio Copilot** and **GitHub assistants** evaluating millions of interactions per week.
- Contributed to an interactive debugger, achieving 3x improvement in bug-resolution rates. Deployed in Visual Studio IDE.

**American Express, AI Labs**

Bangalore, India

*Analyst Intern*

*January 2023 - June 2023*

- Developed a custom Gini gain function that improved (1) regularization on data with high missingness and (2) generalization on multi-market models. Improved an internal version of XGBoost, aimed at credit default/fraud prediction.

**Speech and Language Lab, NTU Singapore** | 

Remote

*Research Intern*

*August 2022 - January 2023*

- Trained convolution-augmented vision transformers to improve the acoustic feature representations of audio signals, improving F1 score our audio analysis pipeline by 10%. Developed a real-time Voice Activity Detection (VAD) system.
- Developed vision transformer models with CRNNs and self-attention to secure top five in DCASE 2023 Challenge Task4b.

## PAPERS AND PUBLICATIONS

C: CONFERENCE, W: WORKSHOP, P: PREPRINT

P1. **Param Biyani, Krishnamurthy Dj Dvijotham** – **SpecGuard: Proving a Task Is Broken Before the Agent Cheats**

*Preprint 2026*

C1. 🏆 **Best Paper Presentation** **Param Biyani\***, Yasharth Bajpai\*, Arjun Radhakrishna, Gustavo Soares, Sumit Gulwani – **RUBICON: Rubric-Based Evaluation of Domain-Specific Human AI Conversations** *AIware 2024*

C2. 🏆 **Best Paper** Y. Bajpai\*, B. Chopra\*, **P. Biyani**, C. Aslan, S. Gulwani, D. Coleman, C. Parnin, A. Radhakrishna, G. Soares – **Let's Fix this Together: Conversational Debugging with GitHub Copilot** *VL/HCC 2024*

C3. Shashank Kirtania, **Param Biyani**, Priyanshu Gupta, Yasharth Bajpai, Roshni Iyer, Sumit Gulwani, Gustavo Soares – **Improving Language Agents Through BREW** *MTI-LLM at NeurIPS 2025; COLM 2026*

W1. **Param Biyani**, Divij Chawla, Elliott Thornley, Christos Ziakas – **ShutdownMod: Turning Any Trajectory into a Shutdown Evaluation** *IAB at NeurIPS 2026*

W2. **Param Biyani**, Shashank Kirtania, Yasharth Bajpai, Sumit Gulwani, Ashish Tiwari – **INDIMATHBENCH: Autoformalizing Mathematical Reasoning Problems with a Human Touch** *Post-AI-FM at AAAI 2026*

W3. B. Chopra\*, Y. Bajpai\*, **P. Biyani**, G. Soares, A. Radhakrishna, C. Parnin, and S. Gulwani – **Exploring Interaction Patterns for Debugging: Enhancing Conversational Capabilities of AI-assistants** *HCI and NLP at NAACL 2024*

## SELECTED RESEARCH PROJECTS

**End-to-end Verified Code Generation in Lean 4** | 📄 June 2026

- Built an end-to-end verified code generation pipeline (code, spec, proof) in Lean 4 evaluated on VERINA benchmark.
- Designed an adversarial tester-generator loop for spec synthesis, with property based testing, reaching 96.3% accuracy.
- Developed a recursive proof agent with dynamic retrieval over Lean 4/Mathlib source reaching 67% proof accuracy.

**IndiMathBench: Formal theorem proving benchmark and autoformalization study** | 📄 Sept 2025

- Created a 312 sized benchmark for Automated Theorem Proving benchmark in Lean4 from Indian Mathematical Olympiads.
- Conducted a systematic study of Human-AI collaboration on formalization tasks.
- Created a VS code extension to speed up Lean annotation by 3.5x, using multi-LLM formalization and group reflection.

**AutoDev: Autonomous Programming Agent, Microsoft** March 2025

- Designed and led a 6+ month research project on automating software development using multi-agent AI systems.
- Built specialized agents for planning, retrieval, code generation, and review, enabling hierarchical task execution.
- Running and quickly iterating on experiments to develop the agent system support and break SOTA on 20+ benchmarks including swe-bench. Directly informed key system-design decisions for GitHub Copilot.

**RUBICON: Human-AI conversation evaluation, Microsoft** | 📄 | 🏢 April 2024

*Under Dr. Gustavo Soares and Dr. Arjun Radhakrishna*

*Microsoft PROSE*

- Devised a learning technique that generates user satisfaction and domain relevant assertions from conversational data.
- Designed a custom scoring function and greedy based algorithm to select top n assertions from a larger pool of generations.

## TECHNICAL SKILLS

|                               |                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------|
| <b>Languages:</b>             | Python, Lean 4, C, C++, Java, C#, JavaScript                                                    |
| <b>AI:</b>                    | Prompt Engineering, RL, Supervised Fine-Tuning, Developing Agentic Systems, RAG, Transformers   |
| <b>Softwares &amp; Tools:</b> | Pytorch, Tensorflow, All popular LLM usage/APIs, Blender 3D, Pandas, Matplotlib, sklearn, Numpy |

## RELEVANT COURSES

|                          |                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Computer Science:</b> | Data Structures and Algorithms, Operating Systems, Object Oriented Programming, Database Management Systems, Computer Networks, Design and Analysis of Algorithms |
| <b>ML/AI:</b>            | Deep Learning, Machine Learning, Reinforcement Learning, Linguistics, Applied Statistical Methods, Cognitive Neuroscience, Computational Learning Theory (Audit)  |
| <b>TAships:</b>          | Machine Learning, Deep Learning, Foundations of Data Science, Object Oriented Programming                                                                         |

## ACHIEVEMENTS

- Best Paper Award, VL/HCC 2024; Awarded by IEEE Symposium on Visual Languages and Human-Centered Computing.
- Best Paper Presentation Award, AIware 2024; Awarded by ACM International Conference on AI-Powered Software.
- Merit Scholarship for being in the **top 3%** of students based on Academic Performance.